

2011
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1–8
- All questions are of equal value

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All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet.

(a) Find $\int x \ln x \, dx$. **2**

(b) Evaluate $\int_0^3 x \sqrt{x+1} \, dx$. **3**

(c) (i) Find real numbers a , b and c such that **2**

$$\frac{1}{x^2(x-1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{c}{x-1}.$$

(ii) Hence, find $\int \frac{1}{x^2(x-1)} \, dx$. **2**

(d) Find $\int \cos^3 \theta \, d\theta$. **3**

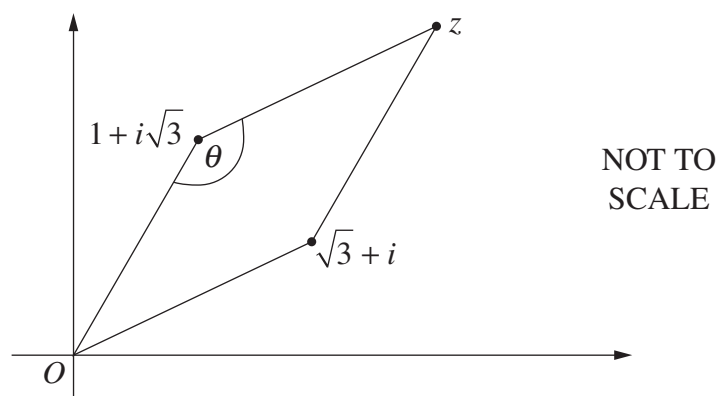
(e) Evaluate $\int_{-1}^1 \frac{1}{5-2t+t^2} \, dt$. **3**

Question 2 (15 marks) Use a SEPARATE writing booklet.

(a) Let $w = 2 - 3i$ and $z = 3 + 4i$.

- (i) Find $\bar{w} + z$. 1
- (ii) Find $|w|$. 1
- (iii) Express $\frac{w}{z}$ in the form $a + ib$, where a and b are real numbers. 2

(b) On the Argand diagram, the complex numbers 0 , $1 + i\sqrt{3}$, $\sqrt{3} + i$ and z form a rhombus.



- (i) Find z in the form $a + ib$, where a and b are real numbers. 1
- (ii) An interior angle, θ , of the rhombus is marked on the diagram. 2
Find the value of θ .

(c) Find, in modulus-argument form, all solutions of $z^3 = 8$. 2

- (d) (i) Use the binomial theorem to expand $(\cos \theta + i \sin \theta)^3$. 1
- (ii) Use de Moivre's theorem and your result from part (i) to prove that 3

$$\cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta.$$

- (iii) Hence, or otherwise, find the smallest positive solution of 2

$$4 \cos^3 \theta - 3 \cos \theta = 1.$$

Question 3 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Draw a one-third page sketch of the graph $y = \sin \frac{\pi}{2}x$ for $0 < x < 4$. **1**

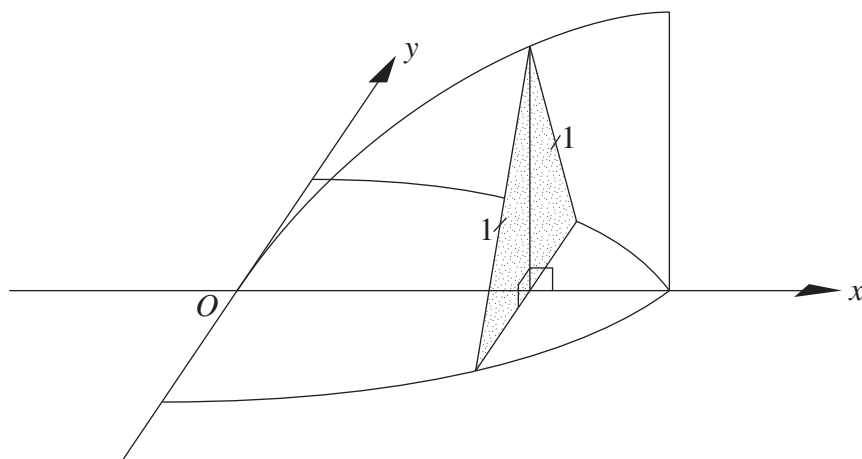
(ii) Find $\lim_{x \rightarrow 0} \frac{x}{\sin \frac{\pi}{2}x}$. **1**

(iii) Draw a one-third page sketch of the graph $y = \frac{x}{\sin \frac{\pi}{2}x}$ for $0 < x < 4$. **2**

(Do NOT calculate the coordinates of any turning points.)

(b) The base of a solid is formed by the area bounded by $y = \cos x$ and $y = -\cos x$ for $0 \leq x \leq \frac{\pi}{2}$. **3**

Vertical cross-sections of the solid taken parallel to the y -axis are in the shape of isosceles triangles with the equal sides of length 1 unit as shown in the diagram.



Find the volume of the solid.

Question 3 continues on page 5

Question 3 (continued)

- (c) Use mathematical induction to prove that $(2n)! \geq 2^n (n!)^2$ for all positive integers n . **3**
- (d) The equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$ represents a hyperbola.
- (i) Find the eccentricity e . **1**
- (ii) Find the coordinates of the foci. **1**
- (iii) State the equations of the asymptotes. **1**
- (iv) Sketch the hyperbola. **1**
- (v) For the general hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, describe the effect on the hyperbola as $e \rightarrow \infty$. **1**

End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

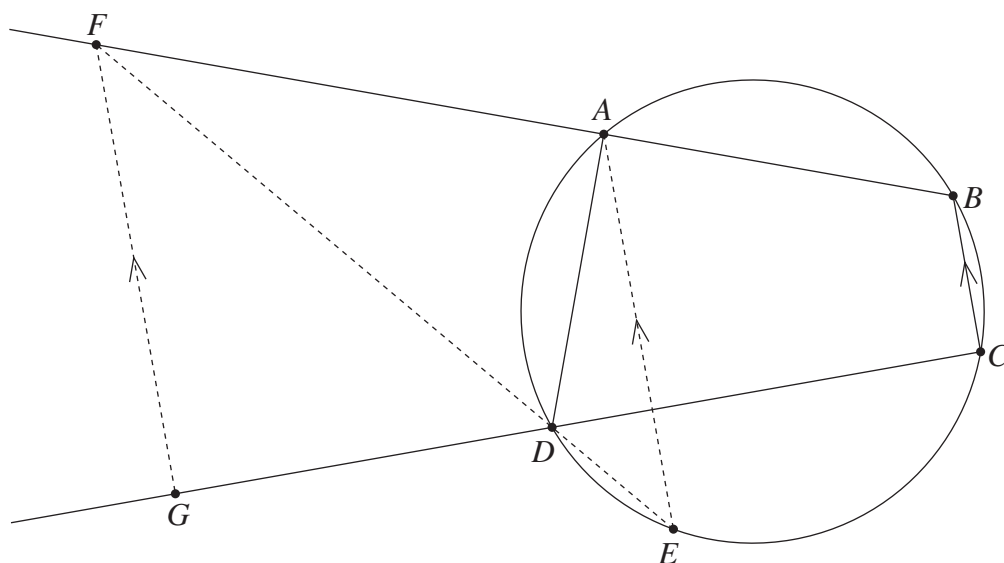
- (a) Let a and b be real numbers with $a \neq b$. Let $z = x + iy$ be a complex number such that

$$|z - a|^2 - |z - b|^2 = 1.$$

- (i) Prove that $x = \frac{a+b}{2} + \frac{1}{2(b-a)}$. 2

- (ii) Hence, describe the locus of all complex numbers z such that $|z - a|^2 - |z - b|^2 = 1$. 1

- (b) In the diagram, $ABCD$ is a cyclic quadrilateral. The point E lies on the circle through the points A, B, C and D such that $AE \parallel BC$. The line ED meets the line BA at the point F . The point G lies on the line CD such that $FG \parallel BC$.



Copy or trace the diagram into your writing booklet.

- (i) Prove that $FADG$ is a cyclic quadrilateral. 2
- (ii) Explain why $\angle GFD = \angle AED$. 1
- (iii) Prove that GA is a tangent to the circle through the points A, B, C and D . 2

Question 4 continues on page 7

Question 4 (continued)

- (c) A mass is attached to a spring and moves in a resistive medium. The motion of the mass satisfies the differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 0,$$

where y is the displacement of the mass at time t .

- (i) Show that, if $y = f(t)$ and $y = g(t)$ are both solutions to the differential equation and A and B are constants, then **2**

$$y = Af(t) + Bg(t)$$

is also a solution.

- (ii) A solution of the differential equation is given by $y = e^{kt}$ for some values of k , where k is a constant. **2**

Show that the only possible values of k are $k = -1$ and $k = -2$.

- (iii) A solution of the differential equation is **3**

$$y = Ae^{-2t} + Be^{-t}.$$

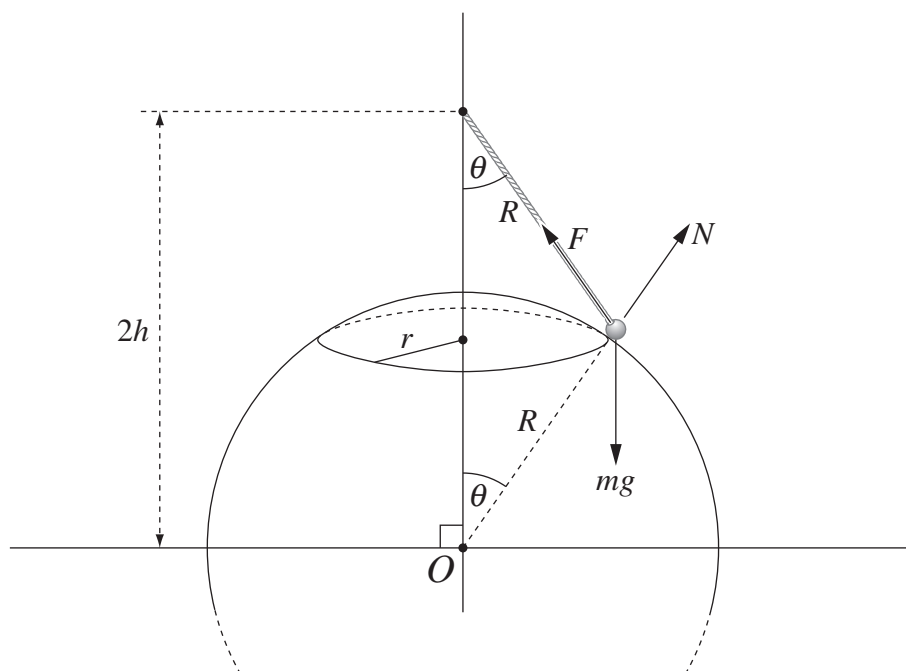
When $t = 0$, it is given that $y = 0$ and $\frac{dy}{dt} = 1$.

Find the values of A and B .

End of Question 4

Question 5 (15 marks) Use a SEPARATE writing booklet.

- (a) A small bead of mass m is attached to one end of a light string of length R . The other end of the string is fixed at height $2h$ above the centre of a sphere of radius R , as shown in the diagram. The bead moves in a circle of radius r on the surface of the sphere and has constant angular velocity $\omega > 0$. The string makes an angle of θ with the vertical.



Three forces act on the bead: the tension force F of the string, the normal reaction force N to the surface of the sphere, and the gravitational force mg .

- (i) By resolving the forces horizontally and vertically on a diagram, show that 2

$$F \sin \theta - N \sin \theta = m\omega^2 r$$

and

$$F \cos \theta + N \cos \theta = mg.$$

- (ii) Show that 2

$$N = \frac{1}{2}mg \sec \theta - \frac{1}{2}m\omega^2 r \operatorname{cosec} \theta.$$

- (iii) Show that the bead remains in contact with the sphere if $\omega \leq \sqrt{\frac{g}{h}}$. 2

Question 5 continues on page 9

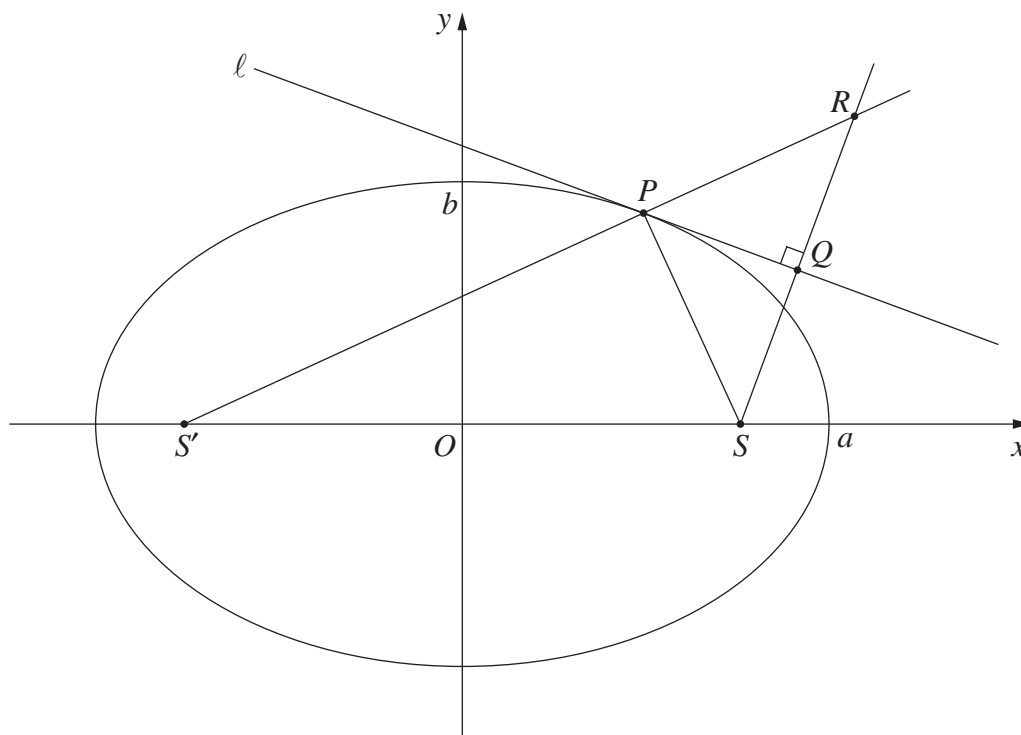
Question 5 (continued)

- (b) If p, q and r are positive real numbers and $p + q \geq r$, prove that

3

$$\frac{p}{1+p} + \frac{q}{1+q} - \frac{r}{1+r} \geq 0.$$

- (c) The diagram shows the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$. The line ℓ is the tangent to the ellipse at the point P . The foci of the ellipse are S and S' . The perpendicular to ℓ through S meets ℓ at the point Q . The lines SQ and $S'P$ meet at the point R .



Copy or trace the diagram into your writing booklet.

- | | |
|--|---|
| (i) Use the reflection property of the ellipse at P to prove that $SQ = RQ$. | 2 |
| (ii) Explain why $S'R = 2a$. | 1 |
| (iii) Hence, or otherwise, prove that Q lies on the circle $x^2 + y^2 = a^2$. | 3 |

End of Question 5

Question 6 (15 marks) Use a SEPARATE writing booklet.

- (a) Jac jumps out of an aeroplane and falls vertically. His velocity at time t after his parachute is opened is given by $v(t)$, where $v(0) = v_0$ and $v(t)$ is positive in the downwards direction. The magnitude of the resistive force provided by the parachute is kv^2 , where k is a positive constant. Let m be Jac's mass and g the acceleration due to gravity. Jac's terminal velocity with the parachute open is v_T .

Jac's equation of motion with the parachute open is

$$m \frac{dv}{dt} = mg - kv^2. \quad (\text{Do NOT prove this.})$$

- (i) Explain why Jac's terminal velocity v_T is given by $\sqrt{\frac{mg}{k}}$. **1**

- (ii) By integrating the equation of motion, show that t and v are related by the equation **3**

$$t = \frac{v_T}{2g} \ln \left[\frac{(v_T + v)(v_T - v_0)}{(v_T - v)(v_T + v_0)} \right].$$

- (iii) Jac's friend Gil also jumps out of the aeroplane and falls vertically. Jac and Gil have the same mass and identical parachutes. **3**

Jac opens his parachute when his speed is $\frac{1}{3}v_T$. Gil opens her parachute when her speed is $3v_T$. Jac's speed increases and Gil's speed decreases, both towards v_T .

Show that in the time taken for Jac's speed to double, Gil's speed has halved.

Question 6 continues on page 11

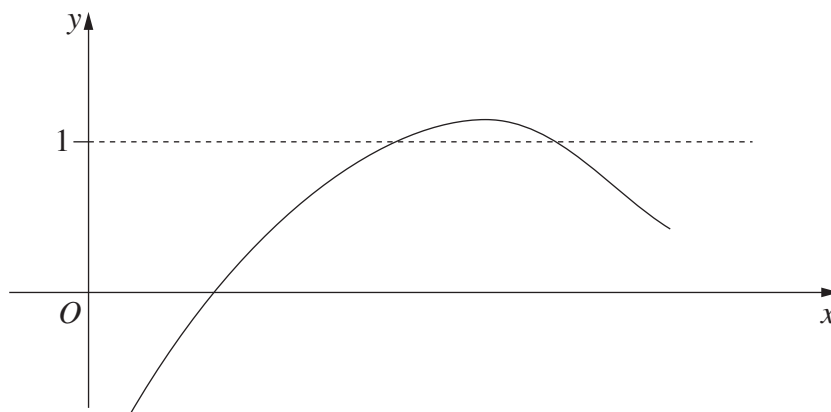
Question 6 (continued)

(b) Let $f(x)$ be a function with a continuous derivative.

(i) Prove that $y = (f(x))^3$ has a stationary point at $x = a$ if $f(a) = 0$ or $f'(a) = 0$. 2

(ii) Without finding $f''(x)$, explain why $y = (f(x))^3$ has a horizontal point of inflexion at $x = a$ if $f(a) = 0$ and $f'(a) \neq 0$. 1

(iii) The diagram shows the graph $y = f(x)$. 3



Copy or trace the diagram into your writing booklet.

On the diagram in your writing booklet, sketch the graph $y = (f(x))^3$, clearly distinguishing it from the graph $y = f(x)$.

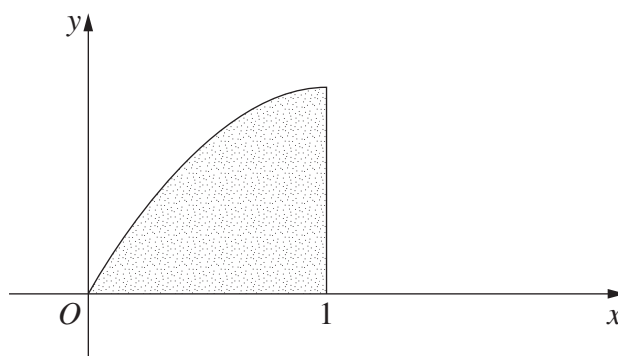
(c) On an Argand diagram, sketch the region described by the inequality 2

$$\left| 1 + \frac{1}{z} \right| \leq 1.$$

End of Question 6

Question 7 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows the graph of $f(x) = \frac{x}{1+x^2}$ for $0 \leq x \leq 1$. **4**



The area bounded by $y = f(x)$, the line $x = 1$ and the x -axis is rotated about the line $x = 1$ to form a solid.

Use the method of cylindrical shells to find the volume of the solid.

(b) Let $I = \int_1^3 \frac{\cos^2\left(\frac{\pi}{8}x\right)}{x(4-x)} dx$.

- (i) Use the substitution $u = 4 - x$ to show that **2**

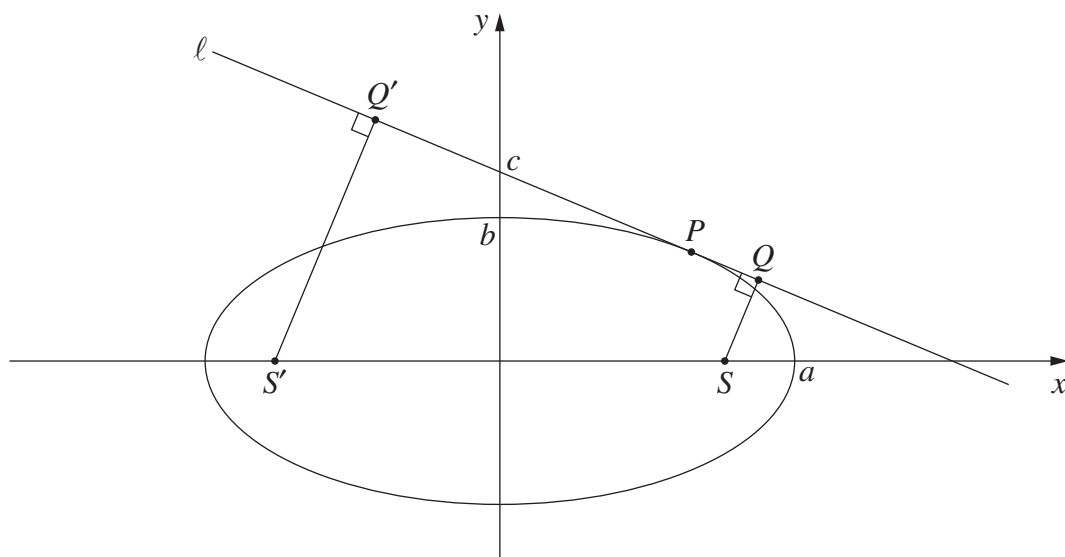
$$I = \int_1^3 \frac{\sin^2\left(\frac{\pi}{8}u\right)}{u(4-u)} du .$$

- (ii) Hence, find the value of I . **3**

Question 7 continues on page 13

Question 7 (continued)

- (c) The diagram shows the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$. Let e be the eccentricity of the ellipse.



The line ℓ is the tangent to the ellipse at the point P . The line ℓ has equation $y = mx + c$, where m is the slope and c is the y -intercept.

The points S and S' are the focal points of the ellipse, where S is on the positive x -axis. The perpendiculars to ℓ through S and S' intersect ℓ at Q and Q' respectively.

- (i) By substituting the equation for ℓ into the equation for the ellipse, show that 3

$$a^2m^2 + b^2 = c^2.$$

- (ii) Show that the perpendicular distance from S to ℓ is given by 1

$$QS = \frac{|mae + c|}{\sqrt{1 + m^2}}.$$

- (iii) It is given that $Q'S' = \frac{|mae - c|}{\sqrt{1 + m^2}}$. 2

Hence, prove that $QS \times Q'S' = b^2$.

End of Question 7

Question 8 (15 marks) Use a SEPARATE writing booklet.

- (a) For every integer $m \geq 0$ let

3

$$I_m = \int_0^1 x^m (x^2 - 1)^5 dx.$$

Prove that for $m \geq 2$

$$I_m = \frac{m-1}{m+11} I_{m-2}.$$

- (b) A bag contains seven balls numbered from 1 to 7. A ball is chosen at random and its number is noted. The ball is then returned to the bag. This is done a total of seven times.

- | | |
|--|----------|
| (i) What is the probability that each ball is selected exactly once? | 1 |
| (ii) What is the probability that at least one ball is not selected? | 1 |
| (iii) What is the probability that exactly one of the balls is not selected? | 2 |

Question 8 continues on page 15

Question 8 (continued)

- (c) Let β be a root of the complex monic polynomial

$$P(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0.$$

Let M be the maximum value of $|a_{n-1}|, |a_{n-2}|, \dots, |a_0|$.

(i) Show that $|\beta|^n \leq M(|\beta|^{n-1} + |\beta|^{n-2} + \cdots + |\beta| + 1)$. **2**

(ii) Hence, show that for any root β of $P(z)$ **3**

$$|\beta| < 1 + M.$$

(d) Let $S(x) = \sum_{k=0}^n c_k \left(x + \frac{1}{x}\right)^k$, where the real numbers c_k satisfy $|c_k| \leq |c_n|$ **3**

for all $k < n$, and $c_n \neq 0$.

Using part (c), or otherwise, show that $S(x) = 0$ has no real solutions.

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2011 HSC Mathematics Extension 2 Marking Guidelines

Question 1 (a)

Criteria	Marks
• Correct solution	2
• Attempts to use integration by parts	1

Question 1 (b)

Criteria	Marks
• Correct solution	3
• Obtains correct primitive, or equivalent merit	2
• Identifies correct substitution	1

Question 1 (c) (i)

Criteria	Marks
• Correct solution	2
• Correctly obtains one constant	1

Question 1 (c) (ii)

Criteria	Marks
• Correct solution	2
• Attempts to integrate and makes some progress	1

Question 1 (d)

Criteria	Marks
• Correct solution	3
• Obtains $\int \cos \theta d\theta - \int \cos \theta \sin^2 \theta d\theta$, or equivalent merit	2
• Attempts to use a relevant trigonometric substitution	1

Question 1 (e)

Criteria	Marks
• Correct solution	3
• Obtains correct primitive, or equivalent merit	2
• Correctly completes the square, or equivalent merit	1

Question 2 (a) (i)

Criteria	Marks
• Correct answer	1

Question 2 (a) (ii)

Criteria	Marks
• Correct answer	1

Question 2 (a) (iii)

Criteria	Marks
• Correct solution	2
• Attempts to multiply by $\frac{3-4i}{3-4i}$, or equivalent merit	1

Question 2 (b) (i)

Criteria	Marks
• Correct answer	1

Question 2 (b) (ii)

Criteria	Marks
• Correct solution	2
• Makes some progress	1

Question 2 (c)

Criteria	Marks
• Correct answer	2
• Attempts to use de Moivre's theorem, or equivalent merit	1

Question 2 (d) (i)

Criteria	Marks
• Correct solution	1

Question 2 (d) (ii)

Criteria	Marks
• Correct proof	3
• Equates real parts	2
• Attempts to use de Moivre's theorem	1

Question 2 (d) (iii)

Criteria	Marks
• Correct solution	2
• Obtains a relevant equation for $\cos 3\theta$	1

Question 3 (a) (i)

Criteria	Marks
• Correct sketch	1

Question 3 (a) (ii)

Criteria	Marks
• Correct limit	1

Question 3 (a) (iii)

Criteria	Marks
• Approximately correct sketch	2
• Correctly indicates ONE key feature of the graph	1

Question 3 (b)

Criteria	Marks
• Correct solution	3
• Obtains a correct primitive for the volume integral, or equivalent merit	2
• Obtains a correct expression for the area of a cross-section, or equivalent merit	1

Question 3 (c)

Criteria	Marks
• Correct proof	3
• Makes substantial progress	2
• Checks that the inequality holds for $n = 1$, or equivalent merit	1

Question 3 (d) (i)

Criteria	Marks
• Correct answer	1

Question 3 (d) (ii)

Criteria	Marks
• Correct answer	1

Question 3 (d) (iii)

Criteria	Marks
• Correct equations	1

Question 3 (d) (iv)

Criteria	Marks
• Correct sketch	1

Question 3 (d) (v)

Criteria	Marks
• Correct answer	1

Question 4 (a) (i)

Criteria	Marks
• Correct solution	2
• Obtains $(x - a)^2 - (x - b)^2 = 1$ or equivalent merit	1

Question 4 (a) (ii)

Criteria	Marks
• Correct answer	1

Question 4 (b) (i)

Criteria	Marks
• Correct proof	2
• Makes some progress	1

Question 4 (b) (ii)

Criteria	Marks
• Correct explanation	1

Question 4 (b) (iii)

Criteria	Marks
• Correct proof	2
• Makes some progress	1

Question 4 (c) (i)

Criteria	Marks
• Correct solution	2
• Differentiates correctly and attempts to substitute, or equivalent merit	1

Question 4 (c) (ii)

Criteria	Marks
• Correct solution	2
• Differentiates correctly and attempts to substitute	1

Question 4 (c) (iii)

Criteria	Marks
• Correct solution	3
• Correctly obtains simultaneous equations for A and B , or equivalent merit	2
• Uses the initial conditions to form an equation relating A and B , or equivalent merit	1

Question 5 (a) (i)

Criteria	Marks
• Correct solution	2
• Resolves forces horizontally or vertically, on a diagram, or equivalent merit	1

Question 5 (a) (ii)

Criteria	Marks
• Correct proof	2
• Attempts to eliminate F	1

Question 5 (a) (iii)

Criteria	Marks
• Correct solution	2
• Attempts to solve $N = 0$, or equivalent merit	1

Question 5 (b)

Criteria	Marks
• Correct proof	3
• Makes some progress	2
• Identifies correct common denominator	1

Question 5 (c) (i)

Criteria	Marks
• Correct solution	2
• Correctly identifies pair of equal angles using reflection property, or equivalent merit	1

Question 5 (c) (ii)

Criteria	Marks
• Correct explanation	1

Question 5 (c) (iii)

Criteria	Marks
• Correct solution	3
• Makes substantial progress	2
• Evidence of proof using similarity $\triangle OSQ$, $\triangle S'SR$, or equivalent merit	1

Question 6 (a) (i)

Criteria	Marks
• Correct explanation	1

Question 6 (a) (ii)

Criteria	Marks
• Correct solution	3
• Obtains correct partial fraction decomposition, or equivalent merit	2
• Correctly separates variables, or equivalent merit	1

Question 6 (a) (iii)

Criteria	Marks
• Correct solution	3
• Correctly evaluates one of the two expressions for time, or equivalent merit	2
• Attempts to substitute relevant speeds into expression from part (ii)	1

Question 6 (b) (i)

Criteria	Marks
• Correct proof	2
• Correctly applies the chain rule for differentiation, or equivalent merit	1

Question 6 (b) (ii)

Criteria	Marks
• Correct explanation	1

Question 6 (b) (iii)

Criteria	Marks
• Correct sketch	3
• Correctly indicates TWO key features of the graph	2
• Correctly indicates ONE key feature of the graph	1

Question 6 (c)

Criteria	Marks
• Correct sketch	2
• Makes some progress	1

Question 7 (a)

Criteria	Marks
• Correct solution	4
• Converts to integrable form, or equivalent merit	3
• Obtains correct integrand, or equivalent merit	2
• Identifies $r = 1 - x$, or equivalent merit	1

Question 7 (b) (i)

Criteria	Marks
• Correct solution	2
• Correctly uses substitution $u = 4 - x$, or equivalent merit	1

Question 7 (b) (ii)

Criteria	Marks
• Correct solution	3
• Obtains correct partial fraction decomposition, or equivalent merit	2
• Adds the two integrals	1

Question 7 (c) (i)

Criteria	Marks
• Correct proof	3
• Obtains correct quadratic equation in x and correct discriminant, or equivalent merit	2
• Obtains correct quadratic in general form or attempt to use $\Delta = 0$, or equivalent merit	1

Question 7 (c) (ii)

Criteria	Marks
• Correct solution	1

Question 7 (c) (iii)

Criteria	Marks
• Correct proof	2
• Writes a correct expression for $QS \times Q'S'$ and attempts to simplify	1

Question 8 (a)

Criteria	Marks
• Correct proof	3
• Makes substantial progress	2
• Uses integration by parts appropriately	1

Question 8 (b) (i)

Criteria	Marks
• Correct answer	1

Question 8 (b) (ii)

Criteria	Marks
• Correct solution	1

Question 8 (b) (iii)

Criteria	Marks
• Correct solution	2
• Recognises one ball is not selected and one ball selected twice, or equivalent merit	1

Question 8 (c) (i)

Criteria	Marks
• Correct solution	2
• Substitutes root of polynomial and attempts to rearrange equation, or equivalent merit	1

Question 8 (c) (ii)

Criteria	Marks
• Correct solution	3
• Makes substantial progress	2
• Correctly sums the geometric series	1

Question 8 (d)

Criteria	Marks
• Correct solution	3
• Shows $\left x + \frac{1}{x}\right \geq 2$, and correctly applies result from part (c) (ii), or equivalent merit	2
• Correctly applies result from part (c) (ii), or equivalent merit	1

Mathematics Extension 2

2011 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
1 (a)	2	4.1	E8
1 (b)	3	4.1	E8
1 (c) (i)	2	4.1	E4, E8
1 (c) (ii)	2	4.1	E8
1 (d)	3	4.1	E8
1 (e)	3	4.1	E8
2 (a) (i)	1	2.1	E3
2 (a) (ii)	1	2.2	E3
2 (a) (iii)	2	2.1	E3
2 (b) (i)	1	2.3	E3
2 (b) (ii)	2	2.2	E3
2 (c)	2	2.4, 7.4	E3, E4
2 (d) (i)	1	2.1, 8.0	E3, E4
2 (d) (ii)	3	2.1, 2.4	E3, E4
2 (d) (iii)	2	8.0	E2
3 (a) (i)	1	1.1	E6
3 (a) (ii)	1	1.5	E2
3 (a) (iii)	2	1.5	E2
3 (b)	3	5.1	E7
3 (c)	3	8.2	E2
3 (d) (i)	1	3.2	E3
3 (d) (ii)	1	3.2	E3
3 (d) (iii)	1	3.2	E3
3 (d) (iv)	1	3.2	E3
3 (d) (v)	1	3.4	E4
4 (a) (i)	2	2.2	E3
4 (a) (ii)	1	2.5	E3
4 (b) (i)	2	8.1	E2
4 (b) (ii)	1	8.1	E2
4 (b) (iii)	2	8.1	E2
4 (c) (i)	2	8.0	E2
4 (c) (ii)	2	8.0	E2
4 (c) (iii)	3	8.0	E2
5 (a) (i)	2	6.3.3	E5

Question	Marks	Content	Syllabus outcomes
5 (a) (ii)	2	6.3.3	E5
5 (a) (iii)	2	6.3.3	E5
5 (b)	3	8.3	E4
5 (c) (i)	2	3.1	E2
5 (c) (ii)	1	3.1	E2
5 (c) (iii)	3	3.1	E2
6 (a) (i)	1	6.2.3	E5
6 (a) (ii)	3	6.2.3	E5, E8
6 (a) (iii)	3	6.2.3	E5
6 (b) (i)	2	8.0	E6
6 (b) (ii)	1	8.0	E6
6 (b) (iii)	3	1.6	E6
6 (c)	2	2.5	E3
7 (a)	4	5.1	E7
7 (b) (i)	2	4.1	E8
7 (b) (ii)	3	4.1	E8
7 (c) (i)	3	3.1	E4
7 (c) (ii)	1	3.1	E4
7 (c) (iii)	2	3.1	E4
8 (a)	3	4.1	E8
8 (b) (i)	1	8.0	E2
8 (b) (ii)	1	8.0	E2
8 (b) (iii)	2	8.0	E2
8 (c) (i)	2	7.4, 8.3	E4, E9
8 (c) (ii)	3	7.4, 8.3	E4, E9
8 (d)	3	7.4, 8.3	E4, E9